Title

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Abstract

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1 Introduction

2 FDG

2.1 Introduction to the FDG model

2.1.1 The FDG model

Consider a group of individuals $V = \{1, 2, \dots, n\}$, where $x_i \in \mathbb{R}$ is the opinion that individual i holds on a given topic. The influences among the individuals are described by a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$: an edge $(i, j) \in E$, with weight $a_{ij} > 0$, means that individual i is influenced by individual j.

The French–Harary–DeGroot (FDG) model describes how each individual, or agent, updates its opinion: the new opinion is a weighted average of the opinions of the agents it is influenced by,

$$x_i(k+1) = \sum_{j=1}^{n} a_{ij}\, x_j(k), \qquad \sum_{j=1}^{n} a_{ij} = 1, \tag{1}$$

or, in vector form, $x(k+1) = Ax(k)$, where A is the weighted adjacency matrix of G (Definition 2.3). The constraint on the weights means that A is row-stochastic, i.e. A is non-negative and $A\mathbf{1}_n = \mathbf{1}_n$.

Theorem 2.1 (Spectral properties of row-stochastic matrices)**.** *If $A \in \mathbb{R}^{n\times n}$ is row-stochastic, then $1 \in \operatorname{spec}(A)$ and $\rho(A) = 1$; hence $\operatorname{spec}(A)$ is contained in the closed unit disk.*

Definition 2.2 (Semi-convergent matrix)**.** A matrix $A \in \mathbb{R}^{n\times n}$ is *semi-convergent* if $\lim_{k\to+\infty} A^k$ exists and is finite.

Since $x(k) = A^k x(0)$, the asymptotic behaviour of the model is determined by the powers of A. Three outcomes are possible:

- *no convergence*: $\lim_{k\to+\infty} x(k)$ does not exist for some $x(0)$; this happens when A is not semi-convergent, e.g. when A has an eigenvalue $\mu \neq 1$ with $|\mu| = 1$;
- *consensus*: $\lim_{k\to+\infty} x(k) = \alpha\mathbf{1}_n$ for some $\alpha \in \mathbb{R}$;
- *disagreement*: $\lim_{k\to+\infty} x(k) = v$ with $v \neq \alpha\mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

2.1.2 Graph-theoretic tools

The outcome of the model depends on the spectrum of A, which in turn depends on the structure of G. We recall the graph notions needed to link the two.

Definition 2.3 (Weighted adjacency matrix)**.** Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ of order n, its *weighted adjacency matrix* is the $n \times n$ matrix A whose (i, j) entry is equal to the edge weight a_{ij} if $(i, j) \in E$, and is equal to zero otherwise.

Definition 2.4 (Strong connectivity and globally reachable nodes)**.** A digraph $G = (V, E)$ is *strongly connected* if there is a directed walk from any node to any other node. A node of G is *globally reachable* if it can be reached from any other node through a directed walk.

With our convention, a directed walk from i to j means that agent i is influenced by agent j, either directly or through intermediaries. Hence G is strongly connected when every agent influences every other agent, at least indirectly, and a globally reachable node is an agent that influences the whole group.

Definition 2.5 (Periodic digraph)**.** A strongly connected digraph is *periodic* if there exists an integer $k > 1$ (the period) that divides the length of every simple cycle of the digraph; otherwise it is *aperiodic*.

Equivalently, a strongly connected digraph G is periodic if and only if the greatest common divisor of the lengths of its simple cycles is larger than 1. In particular, a strongly connected digraph with a self-loop is aperiodic, since the self-loop is a cycle of length 1.

These notions characterize the matrix properties that determine the spectrum of A.

Definition 2.6 (Irreducible and primitive matrices)**.** A non-negative matrix $A \in \mathbb{R}^{n\times n}$, $n \geq 2$, is *irreducible* if $\sum_{k=0}^{n-1} A^k$ is positive, and *primitive* if there exists $k \in \mathbb{N}$ such that A^k is positive.

Theorem 2.7 (Irreducible matrices and digraphs)**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is irreducible if and only if G is strongly connected.*

Theorem 2.8 (Primitive matrices and digraphs)**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is primitive if and only if G is strongly connected and aperiodic. In particular, if G is strongly connected and has at least one self-loop, then A is primitive.*

Definition 2.9 (Strongly connected components and condensation digraph)**.** Given a weighted digraph G, a subgraph S is a *strongly connected component* of G if S is strongly connected and any other subgraph of G strictly containing S is not strongly connected. The *condensation digraph* $C(G)$ is obtained by collapsing each strongly connected component of G into a single node: for any two distinct strongly connected components S_k and S_h, there is an edge (S_k, S_h) in $C(G)$ if and only if there exists an edge $(i, j) \in E$ with $i \in S_k$ and $j \in S_h$ (self-loops are not allowed in $C(G)$). A node of $C(G)$ with no outgoing edges is called a *sink*.

A sink of $C(G)$ is a group of agents that are influenced only by members of the group itself: no agent outside the group can change their opinions.

If G contains a globally reachable node, then $C(G)$ has a unique sink.

2.1.3 Consensus and social power

Theorem 2.10 (Powers of row-stochastic matrices with a simple and strictly dominant eigenvalue)**.** *Let A be row-stochastic. If the eigenvalue 1 is simple and strictly larger in modulus than all the other eigenvalues of A, then A is semi-convergent and*

$$\lim_{k\to+\infty} A^k = \mathbf{1}_n w^\top,$$

where $w \geq 0$ is the dominant left eigenvector of A, normalized so that $w^\top \mathbf{1}_n = 1$. The limit $\mathbf{1}_n w^\top$ is a rank-one projection matrix.

2.1.4 Disagreement

2.1.5 Rosetta stone

The results of this section, together with the consensus and disagreement theorems above, classify all the possible behaviours of the FDG model. The classification is summarized in Table 1, the so-called *Rosetta stone*, which translates each property of the row-stochastic matrix A into a property of the digraph G and into the corresponding asymptotic outcome; for each case, the table also reports an example digraph, its matrix and its spectrum.

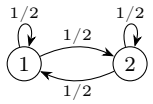
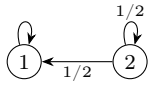
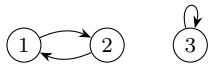
Case	Example digraph	A	spec(A)	Property of A / of G	Outcome
(i)		$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$	$\{1, 0\}$	A primitive / G strongly connected and aperiodic	consensus
(ii)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\{1, -1\}$	A irreducible, not primitive / G strongly connected and periodic	no convergence
(iii)		$\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$	$\{1, 1/2\}$	1 simple and strictly dominant, A not primitive / $C(G)$ has a unique sink, which is aperiodic	consensus
(iv)		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$\{1, 1\}$	1 semi-simple but not simple, strictly dominant / $C(G)$ has at least two sinks, all aperiodic	disagreement
(v)		$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\{1, 1, -1\}$	1 not strictly dominant / G not strongly connected, at least one sink of $C(G)$ is periodic	no convergence

Table 1: Rosetta stone of the FDG model $x(k+1) = Ax(k)$, with A row-stochastic: for each case, an example digraph (unlabelled edges have weight 1), its matrix A, the spectrum of A, the general property of A and of G, and the outcome.

2.2 Stubborness and social power in the FDG model

3 FJ

3.1 Introduction to the FJ model

3.2 Stubborness and social power in the FJ model

4 Conclusions